Emergency Response Guide

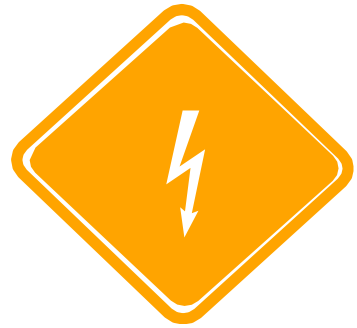
Hino 195 EV / Hino M5 EV / SEA 5e

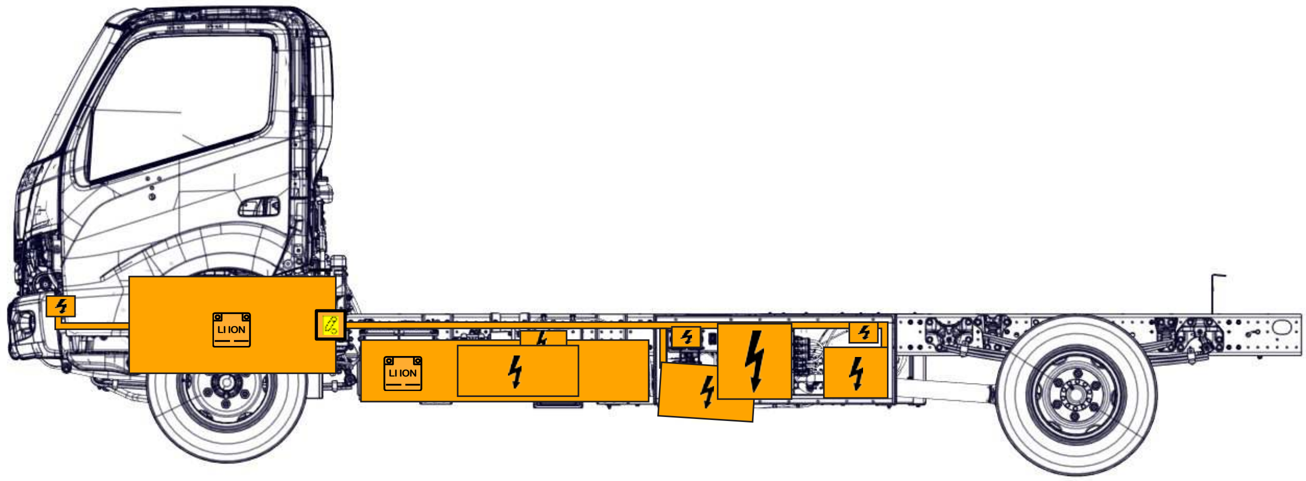
Production Start : 2020

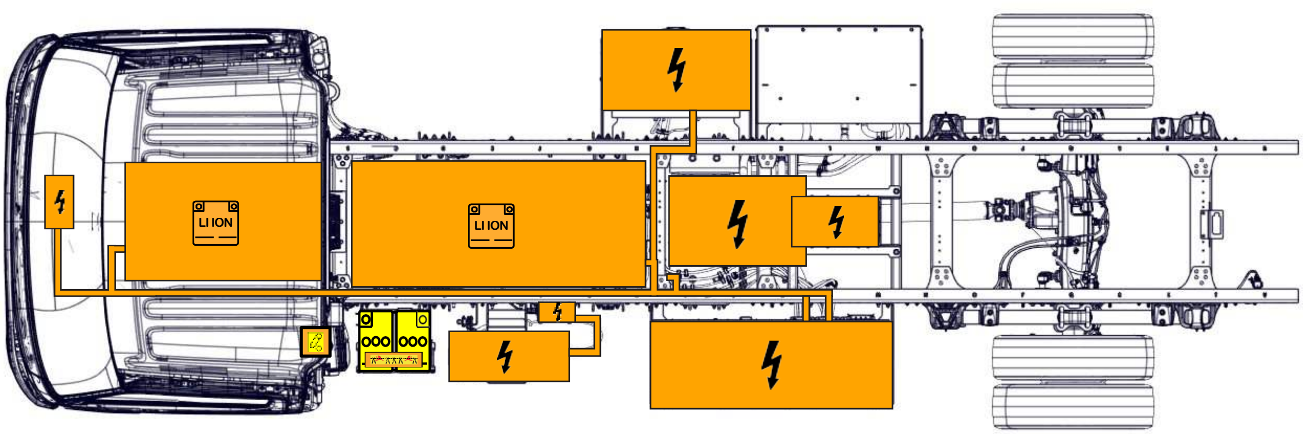
D01266 REV:02 - August 15, 2024

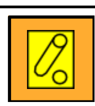
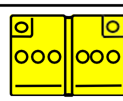
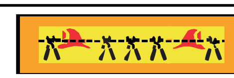
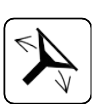
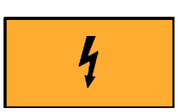
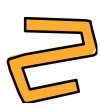
Contents

1. Identification/recognition



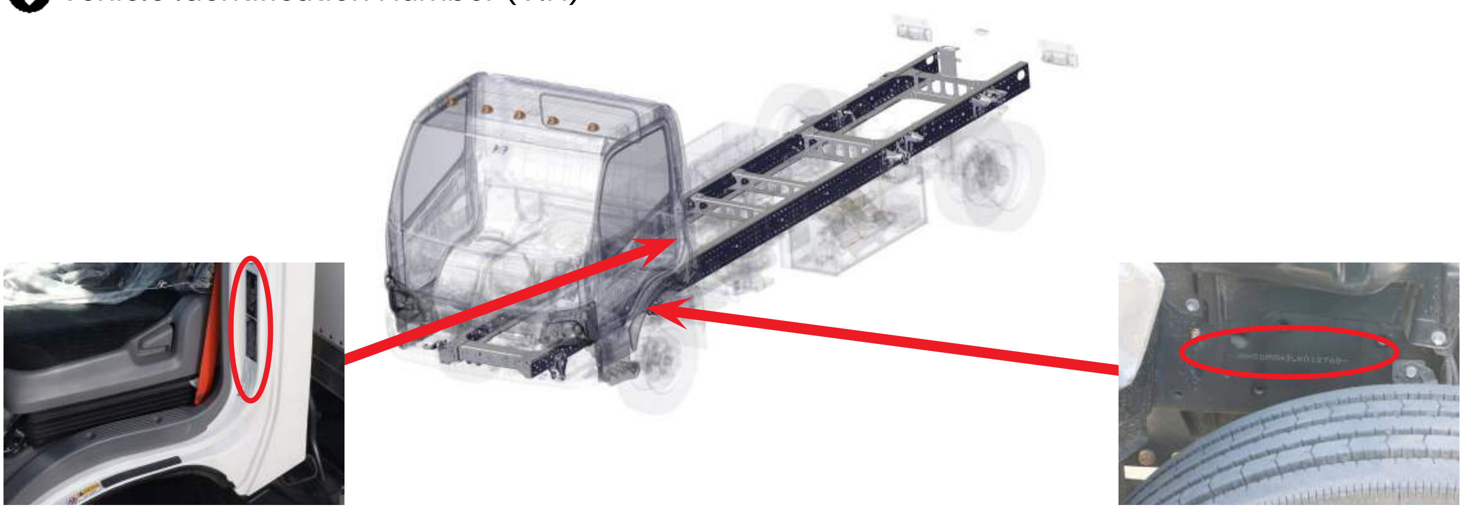




 High voltage lithium-ion battery	 Low voltage device that disconnects the high voltage	 Low voltage battery	 Ignition key	 Seat height adjustment	 Cable cut
 Steering wheel tilt control	 High voltage component	 High voltage cable	 Seat longitudinal adjustment		

POWERED BY
SEA

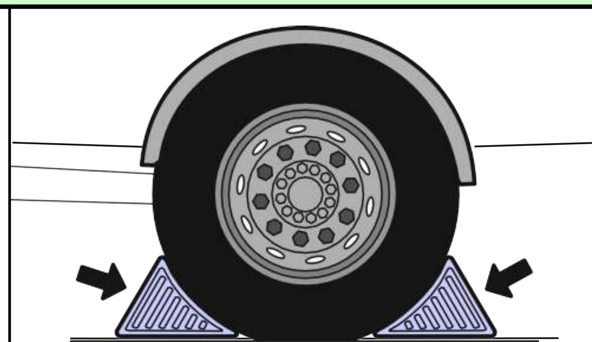
Vehicle Identification Number (VIN)



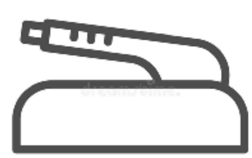
2. Immobilization/stabilization/lifting

Always approach the truck the sides to stay out of the potential travel path. Due to lack of noise, it can be difficult to determine if the truck is running.

1 Chock the wheels.

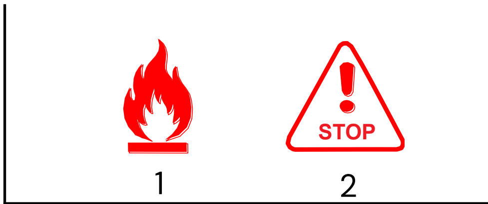


2 Apply the parking brake.

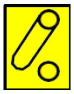


3. Disable direct hazards/safety regulations

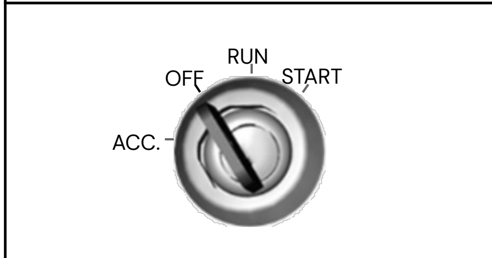
Primary procedure



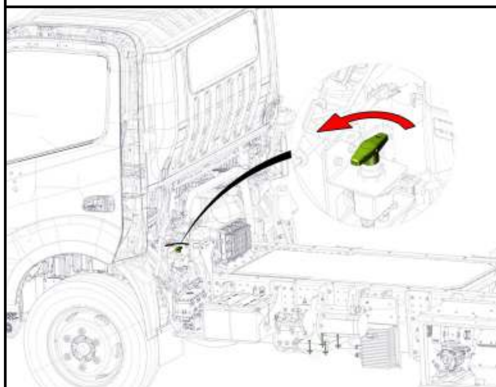
1. Check the instrument cluster for any of the symbols (1) and (2) appearing with a beep sound. If yes, a thermal runaway is detected in the lithium-ion batteries.



2. Turn off the ignition and remove the key.



3. Turn off the chassis switch (up) to initiate the high voltage disconnection process.

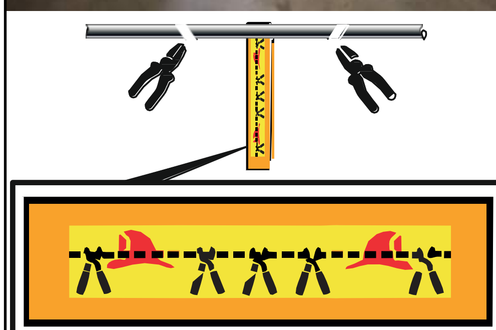


Note:
All the components are designed to discharge their own capacitance within five seconds.

If unable to perform the primary procedure

1. Locate the (2) 12 V batteries behind left side of cab. The (2) batteries are connected with a cable between the positive (+) terminal on one battery to the negative (-) terminal on the other battery to create a 24 V system.

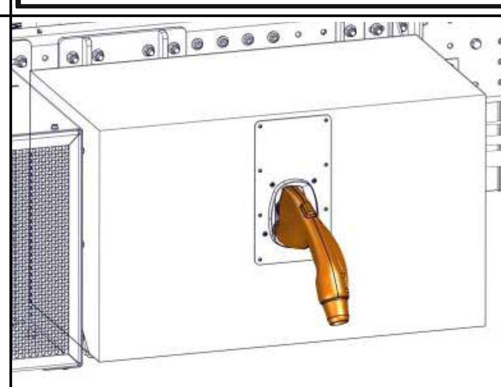
2. Cut the cable between these terminals.



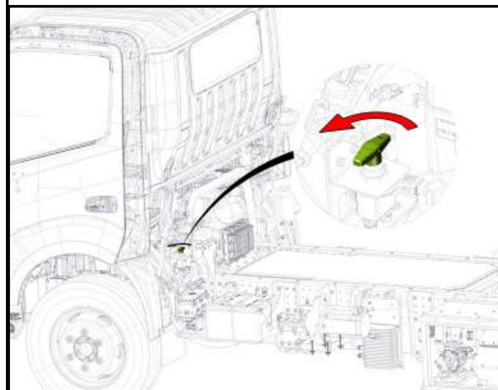
Note:
Cutting this cable shown will disable the traction voltage supply.

If the vehicle is charging:

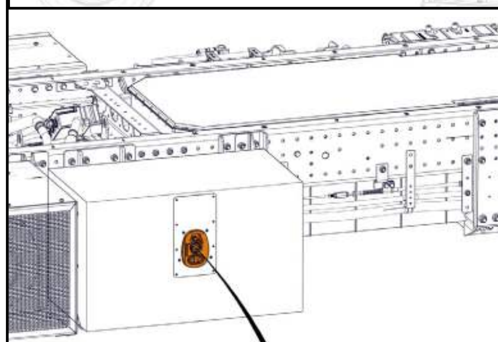
1. Depress button, remove plug from inlet



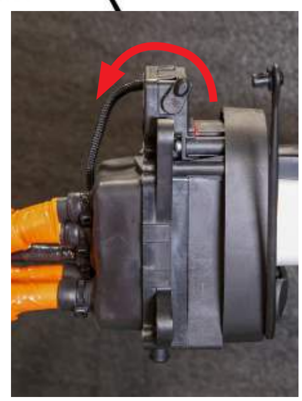
If the charging plug cannot be pulled out: retract the pin manually



1. Turn off the chassis isolator switch (behind cab, drivers side) to initiate the disconnection process.



2. Manual disconnect is located on the top left side of the charger inlet. Reach into back of charger box, and rotate lever towards you to release charger plug.



Note:
All the components are designed to discharge their own capacitance within five seconds.

4. Access to the occupants

Opening doors from the outside:

1. Insert the key (1) in door lock and turn in clockwise direction to unlock the left-hand side door.
2. Insert the key (1) in door lock and turn in counter-clockwise direction to unlock the left-hand side door.
3. To open the door, pull the handle up whilst exerting some force on the door.

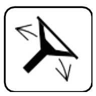
Opening doors from the inside:

1. To open the door, pull the handle (1) whilst exerting some force on the door.

Seat adjustment:

1. To adjust the seat back, lift handle and apply force to seat back.
2. To adjust the seat position, lift up on black handle and slide forward or push backwards.

Steering wheel adjustment:

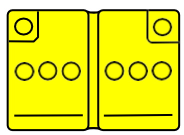
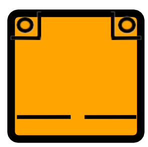


1. To adjust the steering wheel, pull handle towards seat.

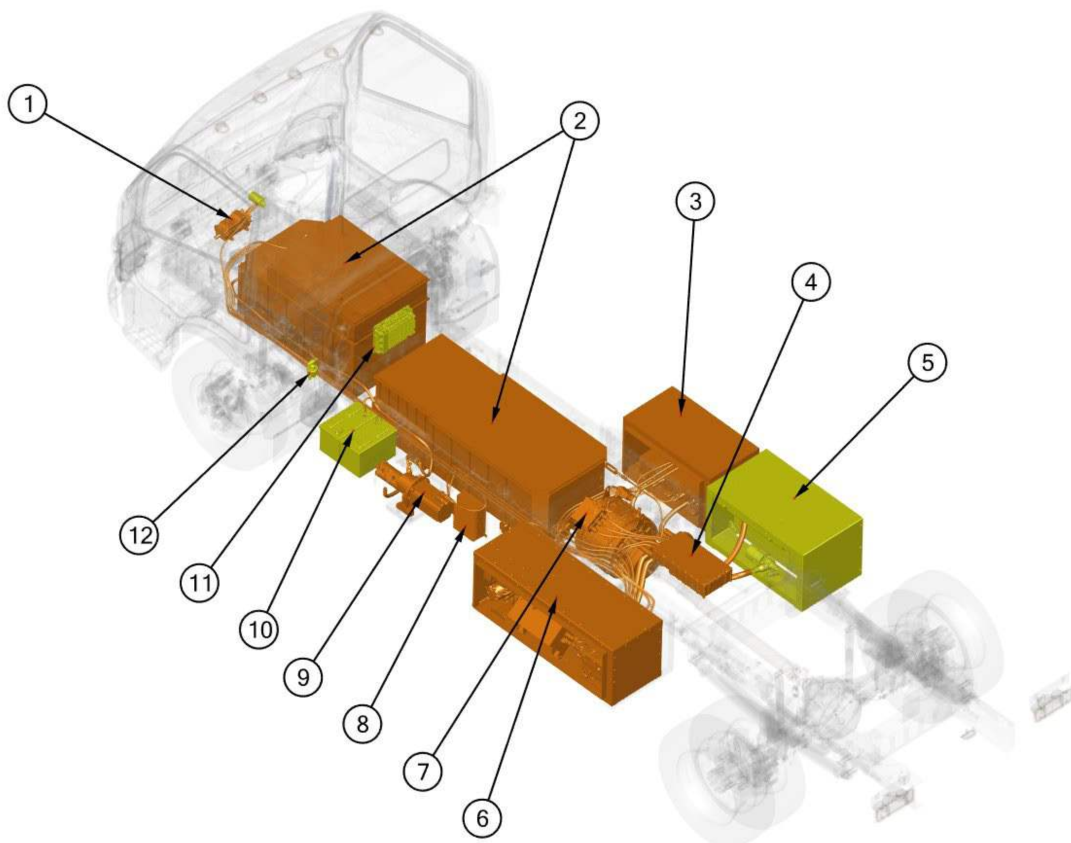
2. Pull the steering up or down in telescopic manner.

3. Steering wheel can rotate towards and away from driver.

5. Stored energy/liquid/gases/solid

		12 V
		400 V

High-voltage component location



1. Heater: Provides heat to the vehicle interior environment.

2. Traction Battery: 2 Li-ion battery pods supply maximum of 400 Volts and 300 Amps. Battery electrodes are made of Carbon, Lithium, Nickel, Manganese and Cobalt.

3. Charging Unit: 19.2kW AC charging. This device is used to convert AC power to DC power.

4. Motor Control Unit (MCU): This device provides HV AC-DC switching to drive the **Traction Motor**.

5. Cooling Box: Provides coolant to the MCU, 3-1 Inverter, motor, and charger.

6. Component Box: Contains Power Distrubtion Unit (PDU), AC Compressor 3-1 Invertor, fuse and relay panel, and the generic Vehicle Control Unit (gVCU).

7. Traction Motor: Provides primary power to the drive train.

8. SEVCON: This device manages the power steering controls.

9. Power Steering Pump: HV electric motor and hydraulic pump assembly provides hydraulic power to the power steering system and braking system.

10. LV Batteries: (2) 12V batteries are connected in series to create a 24V battery system.

11. DC/DC Converter: This device is used to convert 24VDC to 12VDC.

12. LV Battery Disconnect: This switch disables all power to the vehicle, including HV **Traction Batteries.**

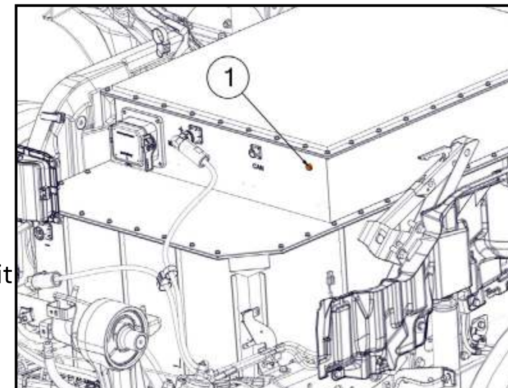
6. In case of fire

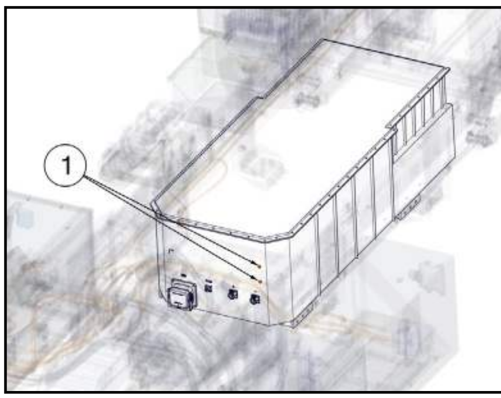
Use a large sustained volume of water to extinguish lithium-ion battery related fire.

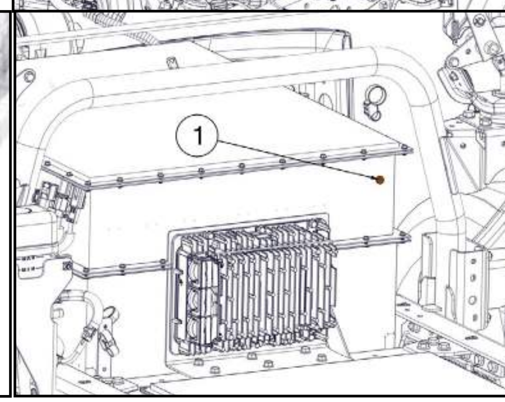
If other materials are involved, use class ABC fire extinguisher.

In case of thermal runaway, the lithium-ion batteries can release hydrogen fluoride.

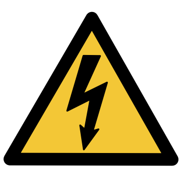
In case of traction battery fire, breather valves (1) may emit large flames as a result of thermal runaway.







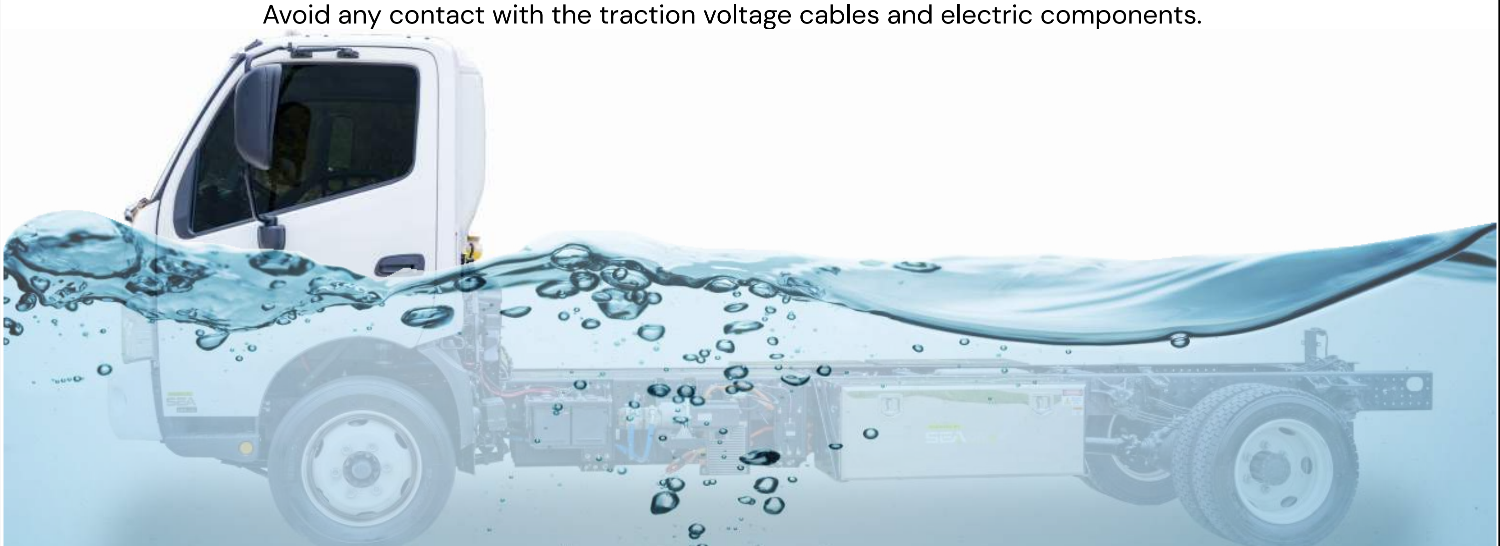
7. In case of water submersion



The damage level of a submerged vehicle may not be visible. Submersion in water can damage 12 V, 24 V and 400 V components.

Handling a submerged truck without appropriate Personal Protective Equipment (PPE) may result in serious injury or death due to electric shock.

Avoid any contact with the traction voltage cables and electric components.



8. Towing/transportation/storage

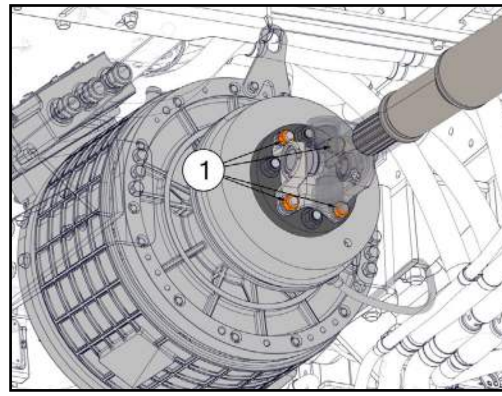
If the traction batteries are damaged, there is a risk of thermal or chemical reaction.

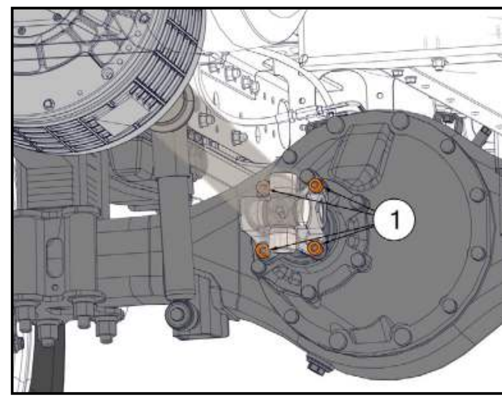
IR

Park the fully electric truck involved in an accident in a suitable place maintaining a safe distance from the other vehicles, buildings and combustible objects.

Risk of delayed fire can happen after the fire suppression of if the lithium-ion batteries are damaged.

Observe the truck for a minimum of 48 hours using thermal infrared camera.





The electric motor can produce electricity when moving the truck with the rear drive tires on the ground. This remains a potential source of electric shock even when the high voltage system is disabled.

Before towing the truck, it is recommended to disconnect the drive shaft from the motor. The drive to the wheels is disabled by uncoupling the propeller shaft by removing the 4 connecting nuts (1) on each end of the shaft coupling.

Maximum loading during lifting and towing:

This information specifies the loading which can be applied when using towing hook, towing hitch cross-member, axles, and torque stay anchorages.

Single towing hook: Do not load the hook more than the vehicle gross weight.

Double towing hooks: Do not load each hook more than half the vehicle gross weight.

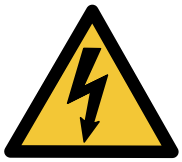
Towing hitch, towing Hitch Cross-Member: Max. 200 mm (7.8 inches) from center of member web

- Lengthways: 20 tons
- Vertically (lift): 7 tons
- Sideways: 17tons

Note:
When the vehicle is towed with the rear suspension lifted, the steering wheel must be locked with steering lock.

Note:
If roof deflectors cannot be removed, tow from the front of the vehicle only.

9. Important additional information

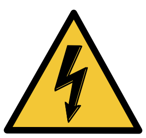
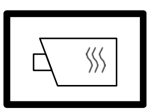


Do not cut any orange cables.

Do not touch any high voltage cables or electric components.

Do not perform any operation on a damaged truck without appropriate Personal Protective Equipment (PPE).

10. Explanation of pictograms used

	Use water to extinguish the fire
	Use ABC powder to extinguish the fire
	General warning sign
	Warning, Electricity
	Use thermal infrared camera
	To indicate the risk of flammability
	To indicate the risk of an explosion
	To indicate the risk of corrosive material/substances
	Hazardous to human health
	To indicate the risk of acute toxicity

Document Approval

DO1266 Rev02
HINO and 5e ERG

Name	Title	Signature
Marco Guedes	VP of Aftermarket and Vehicle Services	Marco Guedes
Jake Woodworth	Mechanical Engineer	Jake Woodworth Jake Woodworth (Aug 22, 2024 06:50 CDT)
Luke Evans	Multimedia Designer	Luke Evans Luke Evans (Aug 26, 2024 08:49 CDT)
Adair Felstad	DDC Manager	Adair Felstad Adair Felstad (Aug 21, 2024 10:09 MDT)